

Narcotic Analgesics

Analgesic: Is the drug that relieves pain, they are divided into two types:

- **Narcotic:** which act centrally as Opioids.
- **Non-narcotic:** they act peripherally as Aspirin.

Narcotics: they relieve pain and also cause drowsiness and sedation. The prototype drug is (Morphine) derived from Opium Poppy.

Opioids: All narcotic analgesics

Opiates (natural opioids): morphine and codeine

Opioids receptors:

Four families of receptors which are: mu, kappa, delta, and sigma. Each of these receptors exhibit different specificity for the drug that they bind.

Distribution of opioid receptors:

CNS (Brain stem, Thalamus, Hypothalamus, Limbic System and Spinal cord).

They are also found in the periphery (like GIT) and in the Immune cells

Mode of action:

All opioids receptors are coupled to inhibitory G-protein (Gi) so, they decrease cAMP level and cause cellular hyperpolarization.

Endogenous Opioids:

These are peptides act as neurotransmitters, found in the complex pain inhibitor system:

1. Enkephalines
2. Dynorphines A & B
3. Endorphines: like β -endorphine

Classification of Opioids:

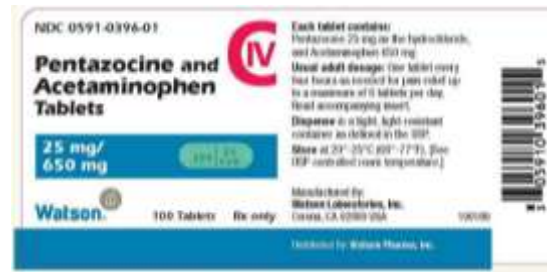
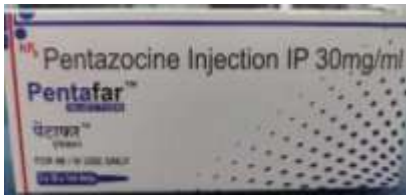
a. Strong agonists: $\propto$ Morphine $\propto$ Meperidine (Pethidine) $\propto$ Methadone $\propto$ Fentanyl $\propto$ Heroin (Diamorphine)



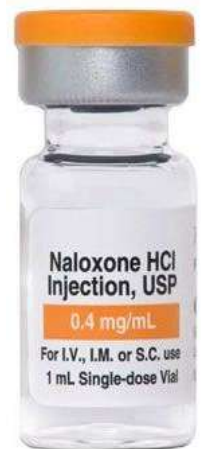
b. Moderate agonists: $\propto$ Codeine $\propto$ Dextropropoxyphene $\propto$ Diphenoxylate $\propto$ Loperamide $\propto$ Dextromethorphan



c. Partial agonist: ⌘ Pentazocine ⌘ Phenazocine



d. Antagonists: ⌘ Naloxone ⌘ Naltrexone ⌘ Nalorphine ⌘ Nalmefene



Morphine

Pharmacological effects:

On CNS: Depressant, stimulant, euphoria and dependence.

A .Depressant effects:

1. Analgesia: by increasing spinal and supraspinal pain threshold at the spinal cord level and altering the brain perception of pain, so the pain is felt but the sensation is not more unpleasant.

2. Drowsiness: morphine has hypnotic and tranquilizer effects.

3. Respiratory depression: it decreases the rate and depth by decreasing the sensitivity of the respiratory center to raised CO₂ tension in the blood (over dose causes respiratory arrest).

Morphine is dangerous in patients with COPD and asthma, in which it will also cause bronchospasm due to histamine release.

4. Inhibition of cough center: (antitussive effect).

5. Increase intracranial pressure: due to CO₂ retention, dilation of cerebral vessels, and increase the CSF pressure, therefore, morphine is contraindicated in patients with severe head injury.

B. Stimulatory actions:

1. Miosis: pin point pupil is characteristic of morphine poisoning and occur as a result of stimulation of mu and kappa receptors. Morphine stimulates 3rd nerve nucleus which cause enhancement of the parasympathetic stimulation to the eye and thus miosis.

There is only little tolerance to this effect (even addicts show pin point pupil). This is an important diagnostic feature of coma because most other causes of coma cause dilation of the pupil.

2. Emesis: 40% nausea and 10% vomiting, this is caused by direct stimulation to the chemoreceptor trigger zone.

On Smooth Muscles:

a. Gastrointestinal tract:

- In the small intestine, decreases motility and causes spasm of sphincters.
- In the large intestine, it decreases peristalsis, and tone is increased. It also decreases intestinal secretions.
- In the stomach, it delays gastric emptying, decreases motility and HCI secretion. It causes constipation and increases the intrasigmoidal pressure, therefore it is contraindicated in patients with diverticular disease and intestinal obstruction.

b. Biliary system: Spasm of sphincter of oddi and increases intrabiliary pressure. Sometimes, biliary colic is worsened by morphine. This effect can be reversed by Naloxone or GTN, but it is not effectively reversed by atropine.

c. The bronchi: bronchospasm.

d. Ureter: spasm.

e. Bladder: spasm of the sphincter that leads to retention of urine in patients with benign prostatic hyperplasia.

f. Uterus: labor is prolonged due to slowing of the effective contraction.

On cardiovascular system:

In the normal persons the effects on CVS are not so important, but can benefit patients with acute left ventricular failure (acute pulmonary edema) due to the following effects:

- Relieve mental distress by tranquilizer effect.
- Relieve cardiac distress by decreasing sympathetic drive.
- Relieve respiratory distress by rendering the respiratory center insensitive to stimuli coming from the congested lungs.

Pharmacokinetics:

*Absorption from GIT is slow.

*It undergoes significant 1st pass elimination, therefore it is usually given IM, IV or SC, also given in buccal and sublingual routes in which the dose is equal to that of the IM route, and it is given as transdermal patch.

*It is also taken by inhalation of crude opioid for the non -medical purposes giving rapid onset of action.

*Morphine enters all tissues and can cross the placenta causing respiratory depression in the fetus at birth.

*Half- life time is 2 hours, duration of action = 4 - 6 hours.

*It is metabolized in liver and kidney to the followings:

- Morphine-6-glucoronide which is active
- Morphine-3-glucoronide which is inactive.

*Conjugated morphine is excreted in urine and small amount appears in the bile.

Clinical Uses:

1. Relief of severe pain, which is either visceral (MI) or somatic (as wounds, fractures). (distinguish between morphine and aspirin is used only in mild and moderate somatic pains).

2. Relief of anxiety in serious and frightening diseases e.g. accidents or hemorrhage (in this case morphine should be given after replacement of fluid loss).

3. Relief of dyspnea in acute left ventricular edema.

4. Production of euphoria and relief pain of dying.

5. Premedication for surgery, in anesthesia (GA & LA).

6. Symptomatic control of non-serious diarrhea e.g. travelers diarrhea (codeine & loperamide are preferred).

7. Suppression of cough (codeine preferred).

Adverse Effects: 1. Respiratory depression. 2. Vomiting. 3. Dysphoria (withdrawal effect). 4. Hypotension. 5. Retention of urine in (BPH). 6. Pruritis (itching around the nose).

Contraindications:

1. **CNS:** a- Head injury because it increases intracranial pressure. b- Acute alcoholism. c- Convulsion disorders.
2. **GIT:** a- Diverticulitis b- Pancreatitis. c- Biliary infections. d- Intestinal obstruction.
3. **Respiratory system:** Asthma and COPD.
4. **In severe liver disease** (acute or chronic) that leads to toxic effects (encephalopathy).
5. **Avoided during labor.**

Tolerance and dependence:

Tolerance occurs mainly to respiratory depression, analgesia, euphoria and sedative effects, but not to the stimulatory actions as miosis. Also tolerance occurs to nausea and vomiting. There is cross tolerance between opioids and other CNS depressants as barbiturates and alcohol.

Treatment of addiction:

1. Withdraw morphine gradually.
2. Substitute with methadone which produces mild withdrawal syndrome, since it has long $t_{1/2}$ (48 hrs).
3. Clonidine decreases the severity of withdrawal syndrome by decreasing the effects of noradrenaline hyperactivity, sometimes; we also use β -blockers as propranolol.

Morphine poisoning (overdose):

Coma, slow and shallow respiration, pinpoint pupil, circulation is maintained till hypoxia is advanced & cyanosis occurs.

Treatment: 1. Support respiration & circulation till antagonize the effect of opioids.
2. Give Naloxone IV it's a pure opioid antagonist, also it is used as a test (diagnostic test) whether the coma is due morphine or not.

Individual opioids

Pethidine (meperidine):

It is commonly used, has less efficacy than morphine, but its pain control beyond reached of codeine. It is like morphine in causing respiratory depression, euphoria & vomiting. Structurally, it is not related to morphine, but it shares many properties with morphine including that of being antagonized by Naloxone.

Pethidine $t_{1/2}$ is 3 hrs, usually given as injection, metabolized to "norpethidine" which can cause excitement and confusions in pethidine poisoning.

Clinical uses of pethidine:

1. Relieve severe pain like post-operative pain.
2. Pre-operative medication.
3. 1st stage of labor

Methadone:

- Structurally related to morphine used orally.
- It causes less sedation and less euphoria than morphine.
- $t_{1/2}$ is 48 hrs and analgesia lasts for 24 hrs.

- Dependence can occur, but is less severe and causes less severe withdrawal syndrome because of long $t_{1/2}$, therefore the addicts on morphine are shifted to methadone as one step of treatment. Main uses: analgesia, cough suppression, treatment of opioids dependence (morphine and heroin).

Diamorphine (heroin):

A semi synthetic drug made by acetylation of morphine. It is the most powerful of all dependence producing drugs.

Fentanyl: used in general anesthesia.

Codeine:

$T_{1/2}$ is 3 hrs. It is methyl morphine, naturally occurring low to moderate efficacy opioids. In the body only 10% is converted to morphine, so its action is 1/10 that of morphine (in same conc. of both). It lacks efficacy in severe pain.



It is used in cough suppression & for short term systemic control of the acute diarrhea. It is usually combined with aspirin or paracetamol.

Dextropropoxyphene:

$T_{1/2}$ is 12 hrs. Analgesic effectiveness is equal to that of codeine. Usually combined with aspirin and paracetamol for greater analgesic effect.

Diphenoxylate and Loperamide: used as antidiarrheal

Dextromethorphan used as cough suppressant

Pentazocine:

- It produces withdrawal syndrome in morphine addicts.
- Can cause psychological & physical dependence.
- Analgesic effect equals that of morphine, but causes less euphoria.
- It's antagonized by Naloxone.

Tramadol:

$T_{1/2}$ is 6 hrs. It binds to μ receptors causing analgesia. It inhibits neuronal noradrenaline uptake &



serotonin release. As effective as pethidine for post-operative

pain and less likely to cause constipation and less addiction and respiratory depression. It may cause convulsions, hallucinations and probably anaphylaxis.

Opioid antagonists:

1. Naloxone:

Pure competitive antagonist that antagonizes both agonists & partial agonists. o Used in the treatment of opioid overdose. It is given IV only because of its high pre-systemic metabolism.

It produces withdrawal syndrome in addicts on opioids.

When given IV, it reverses opioid-induced respiratory depression within 1-2 min and the duration of action is 1hr.

Because opioid analgesic acts much longer than this (1hr), repeated IV doses of Naloxone are required (or IV infusion)

Other use: Diagnosis of opioids overdose (reversal of coma)

To counter the excess effect of opioids after surgical anesthesia and after child birth (given to the baby).

Used in vascular septic shock.

2. Naltrexone:

- It is similar to Naloxone but longer action ($t_{1/2}$ is 4hrs).
- Duration of action 2-3 days because the active metabolites have $t_{1/2}$ of 13 hrs.
- Advantage on Naloxone that it can be given orally.
- It is also used to decrease craving for alcohol in chronic alcoholics.

3. Nalorphine:

Antagonist for all effects of morphine & other opioids. It has also partial agonist effects (i.e. in normal subject it acts as weak agonist, but if given to addict on morphine or other opioids which are stronger, then it acts as antagonists to their effects).

4. Nalmefene:

It is the newest of all these antagonists. It is a derivative of Naltrexone but given only IV. It has a longer $t_{1/2}$ than Naloxone